

9.4.11.

If  $n + 1$  integers are chosen from the set  $\{1, 2, 3, \dots, 2n\}$  where  $n$  is a positive integer, must at least one of them be even? Why?

For any particular positive integer  $n$ , the set  $\{1, 2, 3, \dots, 2n\}$  contains exactly  $2n$  integers. Since  $2n$  is always even, by definition of even  $m = 2k$ , for some integer  $k$ , and  $2n$  is the last number of the given set, then  $2n - 1$  is the largest odd number before  $2n$ , which makes it odd. Then the number of odd integers in the given set is exactly  $n$ . Suppose that all  $n + 1$  chosen integers are odd.

Let  $f : \{n + 1 \text{ chosen integers}\} \rightarrow \{n \text{ odd integers in the set}\}$ .

Since the domain contains  $n + 1$  elements and the codomain contains only  $n$  elements, by the pigeonhole principle  $f$  cannot be injective. But the  $n + 1$  integers were chosen from a set, so they must be distinct. This is a contradiction.

Therefore, not all  $n + 1$  chosen integers can be odd, so at least one of them must be even.

**Solution:** Yes, at least one of the  $n + 1$  chosen integers must be even.